

(a) textbook hard-threshold ABC-PMC: one update per generation

P_{t+1} : the N kept, weighted;
the rejected are discarded

P_t : $(\theta_i, \rho_i, \omega_i^{(t)})$
gen. $< t$ discarded

$\epsilon_{t+1} = \text{quantile}_\alpha \{\rho_i\}_{i \leq N}$
hard: keep $\rho < \epsilon_{t+1}$

$\Sigma_t = 2 \widehat{\text{Cov}}_{\omega^{(t)}}(\{\theta_i\})$
weights carried over

$q_t = \sum_{i \leq N} \omega_i^{(t)} K_{\Sigma_t}(\cdot - \theta_i)$

$\theta^* \sim q_t$, simulate,
keep if $\rho^* < \epsilon_{t+1}$;
one at a time, until N kept

$\omega^{(t+1)} \propto \pi/q_t$
after acceptance

barrier: q_{t+1} needs the complete P_{t+1}

(b) generation-free (ours): one update per arrival

θ^* to a free worker; ρ^* arrives later:
append $(\theta^*, \rho^*, \epsilon_{\tau^*}, \tau^*, w^*)$, $\tau^* = n$

1
state

2
tolerance ϵ

3
weights, Σ

4
proposal

5
draw

6
weight

$\mathcal{H}_n$: all n so far
 $(\theta_i, \rho_i, \epsilon_i, \tau_i, w_i)$

$A_n = \text{Top}_k(\rho)$
 ϵ_n non-increasing, smooth
 K_{ϵ_n}

$\tilde{W}_j \propto w_j K_{\epsilon_n}(\rho_j)$
 $\Sigma_n = s \text{Cov}_{\tilde{W}}(A_n)$

$q_n = \sum_{j \in A_n} \tilde{W}_j K_{\Sigma_n}(\cdot - \theta_j)$

$\theta^* = \theta_J + L_n z$, $J \sim \tilde{W}$
one candidate

$w^* = \pi/\bar{q}_n^{\text{on}}$ at θ^*

$|\mathcal{H}_n| < k$: prior draw, $w^* = 1$,
no ϵ stamp, straight to a worker